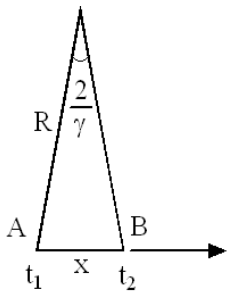
Reference solution 3：

At *P* along the tangent to an electron orbit, an observer sees radiation only briefly. This is the arrival-time interval of light emitted as the velocity direction sweeps the cone angle $2/\gamma$. The electron travels the following distance during that interval:



$$x = \frac{2R}{\gamma} . \qquad (1)$$

Let $\beta = v/c$. The interval between emission at the two points is

$$t_2 - t_1 = \frac{2R}{\gamma c\beta} . \qquad (2)$$

For the tangent observer, subtract the travel time of light from *A* to *B* from this interval, as given below:

$$t' = \frac{2R}{\gamma c} , \qquad (3)$$

Then

$$\Delta T = t_2 - t_1 - t' . \qquad (4)$$

Also,

$$\gamma = \frac{mc^2}{m_0 c^2} = \frac{1}{\sqrt{1-\beta^2}} , \qquad (5)$$

In the limit $1-\beta \to 0$,

$$\frac{2}{\beta}(1-\beta) = \frac{1}{\gamma^2} . \qquad (6)$$

Substitution into (4) gives

$$\Delta T = \frac{R}{c\gamma^3} . \qquad (7)$$

The numerical data give

$$\Delta T = 9.51\times 10^{-20}\text{s} \qquad (8)$$